

Transformers for NLP

- Why attention replaced recurrence
- RNNs process tokens sequentially and can struggle with long-range dependencies.
- Transformers use attention to model token relationships in parallel.
- Goal: understand self-attention, encoder blocks, and fine-tuning.

Self-attention intuition

- Each token attends to other tokens in the sequence.
- Queries, keys, and values produce attention weights.
- Attention weights decide which context tokens matter most.
- Example: disambiguating the word 'bank' from surrounding words.

Transformer encoder block

- Multi-head attention captures different relationship patterns.
- Feed-forward layers transform each token representation.
- Residual connections and normalization stabilize training.
- Stacked encoder blocks build contextual representations.

Fine-tuning and evaluation

- A pretrained Transformer can be adapted to a downstream NLP task.
- Use train, validation, and test splits to evaluate fairly.
- Choose task-appropriate metrics: accuracy, F1, precision, recall.
- Discuss risks: overfitting, data leakage, and biased datasets.